BMS Institute of Technology and Management

(An Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Interdisciplinary -
Project Report on

"SMART CLASSROOMS AND TIME TABLE SCHEDULER FOR HIGHER INSTITUTIONS"

Submitted in partial fulfilment of the requirements for the conferment of degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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VISVESVARAYA TECHNOLOGICAL UNIVERSITY

2026-27

BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(Autonomous Institute under VTU, Belagavi, Karnataka - 590 018)
Yelahanka, Bengaluru, Karnataka – 560119

CERTIFICATE

This is to certify that the **1BPRJ208 – Interdisciplinary Project Work** entitled "**Smart Classrooms and Time Table Scheduler for Higher Institutions"** is a Bonafide work carried out by **Sushant BK (USN:1BY25AI257), Sandesh Naik (USN:1BY25AI212), Rajoth Poddar(USN:1BY25AI186), Rohan H (USN:1BY25AI197),** in partial fulfillment for the award of Bachelor **of Engineering Degree in Artificial Intelligence and Machine Learning** of the **Visvesvaraya Technological University, Belagavi** during the year 2026-2027. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in this report. The Interdisciplinary Project report has been approved as it satisfies the academic requirements in respect of Interdisciplinary Project work for B.E. Degree.

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DECLARATION

We, hereby declare that the Interdisciplinary Project work (1BPRJ208) titled " **Smart Classrooms and Time Table Scheduler for Higher Institutions**" is a record of original Interdisciplinary Project work undertaken for partial fulfilment of Bachelor of Engineering in Artificial Intelligence and Machine Learning of the Visvesvaraya Technological University, Belagavi during the year 2026-27. We have completed this Interdisciplinary Project work under the guidance of **Prof. Umesh T, Assistant Professor.**

I also declare that this Interdisciplinary Project report has not been submitted for the award of any degree, diploma, associate ship, fellowship or other titles anywhere else.

Date:18/06/2026

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ABSTRACT

Smart Classrooms and Time Table Scheduler for Higher Institutions is an intelligent web-based system designed to automate classroom allocation and timetable generation. The system optimizes the scheduling process by considering faculty availability, classroom capacity, subject requirements, and institutional constraints. It minimizes scheduling conflicts, reduces manual effort, and improves resource utilization. The platform provides separate interfaces for administrators, faculty, and students, enabling efficient timetable management and real-time updates. By integrating automation and smart scheduling techniques, the proposed system enhances academic planning, supports effective classroom management, and contributes to a more organized and productive learning environment in higher educational institutions.

ACKNOWLEDGEMENT

We are happy to write a Interdisciplinary Project report after completing it successfully. This Interdisciplinary Project would not have been possible without the guidance, assistance and suggestions of many individuals. We would like to express our deep sense of gratitude to each and everyone who has helped us to make this Interdisciplinary Project a success.

We heartily thank **Dr. Sanjay H A, Principal, BMS Institute of Technology & Management** for his constant encouragement and inspiration in taking up this Interdisciplinary Project.

We heartily thank **Dr. Anupama H S, Head of the Department and Dr. Pradeep K R, Associate Professor & Cluster Head-AIML-2, Artificial Intelligence and Machine Learning, BMS Institute of Technology & Management** for their constant encouragement and inspiration in taking up this mini-project.

We gratefully thank our Interdisciplinary Project guides**, Mr. Umesh T, Asst. Professor, Dept. of Artificial Intelligence and Machine Learning,** for his/her encouragement and advice throughout the course of the Interdisciplinary Project work.

We heartily thank Interdisciplinary Project coordinators **Dr. Chidananda K, Prof. Manasa K Assistant Professor and Prof. Lakshmi M R Assistant Professor Dept. of Artificial Intelligence and Machine Learning,** for their constant follow-up and advice throughout the Interdisciplinary Project work.

Special thanks to all the staff members of the **Artificial Intelligence and Machine Learning** Department for their help and kind co-operation.

Lastly, we thank our parents and friends for their encouragement and support given to us in order to finish this Interdisciplinary Project work.

By,

Sushant BK

Sandesh Naik

Rajoth Poddar

Rohan H

VISION AND MISSION OF THE DEPARTMENT

VISION

To emerge as a leading department in AI and ML by preparing skilled, responsible, and eco-friendly professionals who use technology to improve society.

MISSION

1. To equip students with in-depth knowledge in Artificial Intelligence and Machine Learning, built upon a robust foundation in Computer Science and Engineering through industry focused curriculum with practical learning.
2. To encourage research and innovation through industry partnerships and sustainable technology practices.
3. To inspire students toward ethical leadership and entrepreneurship through innovative and collaborative student-led activities.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

1. Possess essential professional engineering skills that make them confident to develop high-quality AI solutions for various application domains under realistic constraints.
2. Demonstrate the importance of life-long learning through professional development, computing practises, and specialized certifications.
3. Engage and succeed in their professional careers through team work, ethical behaviour, proactive involvement, and effective communication.

PROGRAM SPECIFIC OUTCOMES (PSOs)

1. Ability to apply acquired skills to build optimized solutions adhering to principles and practices of Computational Intelligence.
2. Employ ethical strategies and policies in project and product development.

PROGRAM OUTCOMES (POs)

PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Sustainable Development Goals (SDG) Mapping



This project, Sanket, a sign-to-speech translator and a speech-to-sign translator, is a significant application in meeting the goals of the United Nations Sustainable Development Goals (SDGs). The project is a significant application with respect to meeting goals, which are clearly described in the following points:

SDG 4: Quality Education

Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

The Smart Classroom System enhances the teaching and learning process by integrating modern technologies into education. It provides an interactive and engaging learning environment, improves access to educational resources, and supports effective communication between teachers and students, thereby contributing to better learning outcomes.

SDG 9: Industry, Innovation and Infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

The Smart Classroom System utilizes advanced technologies such as Artificial Intelligence (AI) and the Internet of Things (IoT) to modernize educational infrastructure. By promoting innovation and digital transformation in classrooms, the system helps create a technology-driven learning environment that supports efficient and intelligent educational practices.

SDG 11: Sustainable Cities and Communities

Ensure cities and communities are inclusive, safe, resilient, and sustainable.

The Automated Timetable Scheduling System helps educational institutions use classrooms, laboratories, and faculty resources efficiently. By preventing scheduling conflicts and improving resource management, the system creates a well-organized and sustainable learning environment for students and staff.

SDG 12: Responsible Consumption and Production

Ensure sustainable consumption and production patterns.

The Automated Timetable Scheduling System reduces the wastage of resources by optimizing class schedules and classroom allocation. The system ensures that available facilities are used effectively, improving operational efficiency and promoting responsible resource utilization within educational institutions.

TABLE OF CONTENTS

ABSTRACT **II**

1. INTRODUCTION **01**

1.1. Background 01
1.2. Problem Statement 01
1.3. Significance of the Project 01
1.4. Project Context 02
1.5. Objectives 02
1.6. Scope of the Project 02
1.7. Organization of the Report 03

2. LITERATURE REVIEW **04**

2.1. Introduction 04
2.2. Overview of Timetable Scheduling Systems 04
2.3. Existing Approaches to Timetable Scheduling 04
2.3.1. Manual Timetable Scheduling 04
2.3.2. Rule-Based Scheduling System 04
2.3.3. Optimization-Based Scheduling Systems 05
2.4. Review of Related Studies 05
2.5. Technologies Used in Existing System 06
2.6. Research Gap 06
2.7. Summary 07

3. SYSTEM DESIGN **08**

3.1. Introduction 08
3.2. System Architecture 08
3.3. System Requirements 09
3.4. Methodology 09
3.5. Flowchart 10
3.6. Data Flow Diagram 11
3.7. Module Description 12
3.8. User Interface Design 12
3.9. Codes 16
3.10. Implementation 23
3.11. Summary 23

4. RESULT AND DISCUSSION **24**

4.1. Introduction 24
4.2. Result Analysis 24
4.3. Discussion 25

4.4. Validation 26
4.5. Performance Evaluation 26
4.6. Summary 27

5. CONCLUSION 28

5.1. Introduction 28
5.2. Summary of Findings 28
5.3. Achievements of the Project 29
5.4. Benefits of the Proposed System 29
5.5. Limitations of the System 30
5.6. Recommendations 30
5.7. Future Scope 31
5.8. Conclusion 31

REFERENCES

PLAGIARISM REPORT

LIST OF FIGURES

Figure No.	Figure Name	Page No.
3.1	Architecture of Proposed System	8
3.2	System Flowchart	10
3.3	Level 0 DFD Flowchart	11
3.4	Level 1 DFD Flowchart	11
3.5	Dashboard View	13
3.6	Class Entry View	13
3.7	Subject Entry View	14
3.8	Teacher Entry View	14
3.9	Generated Timetable	15
3.10 - 3.23	Codes	16 - 22

LIST OF TABLES

Table No	**Table Name**	**Page No.**
3.1	Hardware Requirements of the Proposed System	**9**
3.2	Software Requirements of the Proposed System	**9**
4.1	Test Case	25
4.2	Comparison Between Manual Timetable Preparation and Automated Timetable Scheduler System	26

LIST OF ABBREVIATIONS

Abbreviation	Meaning	Page No.
AIML	Artificial Intelligence and Machine Learning	**01,33**
GUI	Graphical User Interface	2,6
SOTT	Stochastic Optimization Timetabling Tool	5
RAM	Random Access Memory	9
IDE	Integrated Development Environment	9
DFD	Data Flow Diagram	11
MyQSL	My Structured Query Language	30
SQLite	Structured Query Language Lite	30
AI	Artificial Intelligence	30,31
IoT	Internet of Things	33
LMS	Learning Management System	33
ICECA	International Conference on Electronics, Communication and Aerospace Technology	33

CHAPTER-1

INTRODUCTION

1.1 Background

In educational institutions, preparing timetables is an essential administrative task that ensures the effective allocation of subjects across available time slots. A well-structured timetable helps maintain academic discipline and ensures the smooth conduct of classes. Traditionally, timetable preparation is performed manually, requiring significant time and effort from faculty members and administrators.

Manual timetable generation often involves managing multiple constraints such as subject allocation, available periods, and scheduling conflicts. As the number of subjects increases, creating and modifying timetables becomes increasingly complex. Human errors such as overlapping schedules, uneven distribution of classes, and inefficient use of available periods are common in manual systems.

With the advancement of information technology, automated scheduling systems have emerged as effective solutions to overcome these challenges. The Automated Timetable Scheduler System is designed to simplify the timetable creation process by generating schedules automatically based on predefined subject information. The system minimizes human intervention and improves the efficiency and accuracy of timetable generation.

1.2 Problem Statement

Timetable preparation is a critical task in educational institutions. Traditional manual methods are often time-consuming, labor-intensive, and prone to errors. Creating schedules manually becomes difficult when handling multiple subjects and limited time slots. Modifying an existing timetable due to changes in academic requirements further increases the complexity of the process.

The lack of automation can result in scheduling conflicts, inefficient utilization of available periods, and increased workload for administrative staff. Therefore, there is a need for an automated system that can generate timetables efficiently while reducing manual effort and ensuring conflict-free scheduling.

1.3 Significance of the Project

The Automated Timetable Scheduler System provides a practical solution for educational institutions by automating the timetable creation process. The system reduces the time required

to prepare schedules and minimizes the occurrence of human errors. By generating timetables automatically, administrators can focus on other academic and management activities.

The project contributes to improved efficiency, better organization of academic activities, and effective utilization of available time slots. It also provides flexibility in generating and modifying timetables whenever necessary. The user-friendly interface makes the system accessible to users with minimal technical knowledge.

1.4 Project Context

The Automated Timetable Scheduler System is developed to assist schools, colleges, and educational institutions in generating subject-wise timetables. The system accepts subject details as input and automatically creates a timetable based on the available scheduling structure.

The project is implemented using Python programming language and provides a graphical user interface for easy interaction. The generated timetable can be viewed and stored for future reference. The system focuses solely on timetable generation and does not include additional functionalities such as attendance management or student performance tracking.

1.5 Objectives

The main objectives of the Automated Timetable Scheduler System are:

1. To develop an automated system for timetable generation.
2. To generate conflict-free schedules for different subjects.
3. To reduce the manual effort involved in timetable preparation.
4. To save time and resources required for academic scheduling.

1.6 Scope of the Project

The scope of the project is limited to the generation of academic timetables based on subject information provided by the user. The system allows users to enter subject details and automatically generate a timetable without scheduling conflicts.

The project includes:

- Subject information management.
- Automatic timetable generation.
- Display of generated timetables.
- Storage of timetable data.

The project does not include:

- Attendance management.
- Student record management.
- Examination scheduling.
- Faculty performance monitoring.
- Online timetable sharing.

1.7 Organization of the Report

This report is organized into five chapters. Chapter 1 presents the introduction, background, objectives, significance, and scope of the project. Chapter 2 provides a review of existing literature, technologies, and methodologies related to timetable scheduling systems. Chapter 3 explains the system design, architecture, methodology, and implementation details. Chapter 4 presents the results obtained from the system along with discussions and validation of the outcomes. Chapter 5 concludes the project by summarizing the findings, achievements, limitations, and future scope of improvements.

CHAPTER-2

LITERATURE REVIEW

2.1 Introduction

A literature review provides an understanding of the existing methods, technologies, and research contributions related to timetable scheduling systems. Timetable generation is a complex scheduling problem that involves allocating subjects, classrooms, and time slots while satisfying various constraints. Over the years, researchers have proposed several approaches to automate timetable generation and improve scheduling efficiency. This chapter reviews existing timetable scheduling systems, optimization techniques, and relevant research studies that have contributed to the development of automated scheduling solutions.

2.2 Overview of Timetable Scheduling Systems

Timetable scheduling is an important activity in educational institutions as it ensures the effective organization of academic activities. Traditionally, timetables were prepared manually by academic administrators. However, manual scheduling often leads to conflicts, inefficient allocation of resources, and increased administrative workload.

To overcome these challenges, computerized timetable scheduling systems have been developed. These systems use algorithms and optimization techniques to generate schedules automatically while considering predefined constraints. Modern scheduling systems reduce human effort, improve accuracy, and provide flexibility in timetable management.

2.3 Existing Approaches to Timetable Scheduling

2.3.1 Manual Timetable Scheduling

Manual timetable preparation involves assigning subjects to available time slots using paper records or spreadsheet applications. Although this approach is simple to implement, it becomes increasingly difficult as the number of subjects and scheduling constraints grows. Manual systems are time-consuming and highly susceptible to human errors.

2.3.2 Rule-Based Scheduling Systems

Rule-based scheduling systems generate timetables according to predefined rules and constraints. These systems ensure that specific scheduling conditions are satisfied during timetable generation. While rule-based systems improve accuracy compared to manual methods, they may struggle to handle complex scheduling scenarios involving numerous constraints.

2.3.3 Optimization-Based Scheduling Systems

Optimization techniques have been widely applied to timetable scheduling problems. These methods attempt to generate optimal or near-optimal schedules while minimizing conflicts and maximizing resource utilization. Common optimization techniques include Genetic Algorithms, Simulated Annealing, Constraint Programming, Integer Programming, and Ant Colony Optimization.

2.4 Review of Related Research Studies

Study 1: Genetic Algorithm for University Course Timetabling

Yu and Sung (2002) proposed a Genetic Algorithm-based approach for university course timetabling. The study demonstrated that Genetic Algorithms are effective in handling complex scheduling constraints and generating feasible timetables within a reasonable time. The researchers concluded that evolutionary algorithms can produce high-quality schedules for educational institutions.

Study 2: Stochastic Optimization Timetabling Tool

Abramson and Abela developed the Stochastic Optimization Timetabling Tool (SOTT), which incorporates Genetic Algorithms and Simulated Annealing techniques for university timetable generation. The study emphasized the importance of satisfying both hard constraints and soft constraints while generating schedules. The proposed system successfully minimized timetable conflicts and improved scheduling efficiency.

Study 3: Evolutionary Approaches for Timetable Scheduling

Doulaty et al. (2013) presented an evolutionary approach for solving university timetabling problems. The researchers highlighted that timetable generation is an NP-complete problem and requires heuristic techniques for practical implementation. Their work demonstrated that Genetic Algorithm-based approaches can generate efficient schedules within acceptable computational time.

Study 4: Review of Genetic Algorithm-Based Timetable Systems

Nathaniel et al. (2019) reviewed various timetable scheduling systems that utilize Genetic Algorithms. The study concluded that automated scheduling systems significantly reduce timetable generation time and improve scheduling accuracy compared to manual methods. The researchers also noted that Genetic Algorithms are among the most widely adopted techniques for solving educational timetabling problems.

Study 5: Optimization Algorithms for University Timetabling

Alghamdi et al. (2020) reviewed multiple optimization algorithms used in university timetable scheduling, including Genetic Algorithms, Particle Swarm Optimization, and other heuristic techniques. The study found that optimization algorithms improve resource allocation and reduce scheduling conflicts while satisfying institutional constraints.

Study 6: Hyper-Heuristic Approaches in Educational Timetabling

Pillay (2016) reviewed the application of hyper-heuristic techniques in educational timetabling. The study showed that hyper-heuristics provide generalized solutions capable of addressing different types of scheduling problems. These approaches have gained popularity because of their flexibility and adaptability to varying scheduling requirements.

2.5 Technologies Used in Existing Systems

Python Programming Language

Python has become one of the most popular programming languages for developing scheduling systems due to its simplicity, readability, and extensive library support. Python enables rapid application development and supports the implementation of various scheduling algorithms.

Graphical User Interfaces

Many timetable scheduling systems utilize graphical user interfaces (GUIs) to simplify user interaction. GUI frameworks such as Tkinter allow users to enter scheduling information, generate timetables, and view results efficiently.

Data Storage Techniques

Scheduling systems commonly use databases or file-based storage mechanisms to maintain subject information and generated timetables. These storage techniques ensure that scheduling data can be accessed and modified when required.

2.6 Research Gap

The literature review reveals that most existing timetable scheduling systems employ advanced optimization algorithms such as Genetic Algorithms, Simulated Annealing, and Hyper-Heuristics. While these techniques are highly effective, they often increase system complexity and computational requirements.

Many available solutions are designed for large universities and include extensive functionalities such as classroom allocation, faculty management, and examination scheduling. Such systems may be difficult to implement and maintain for smaller educational institutions.

Therefore, there is a need for a simple, user-friendly, and cost-effective timetable generation system that focuses primarily on subject scheduling. The proposed Automated Timetable

Scheduler System addresses this gap by providing a lightweight solution that automatically generates conflict-free timetables without requiring complex optimization techniques or extensive computational resources.

2.7 Summary

This chapter reviewed the existing literature related to timetable scheduling systems and optimization techniques. Various research studies have demonstrated the effectiveness of automated scheduling approaches in reducing conflicts and improving efficiency. Technologies such as Python and graphical user interfaces have contributed significantly to the development of user-friendly scheduling systems. The review also identified a research gap in the availability of simple and lightweight timetable generation solutions for educational institutions. The proposed system aims to address this gap by providing an efficient and easy-to-use timetable scheduling application.

CHAPTER-3

SYSTEM DESIGN

3.1 Introduction

The Automated Timetable Scheduler System is designed to simplify the process of timetable generation in educational institutions. The system accepts subject information from the user and automatically generates a timetable based on predefined scheduling rules. The application is developed using Python and Tkinter, providing a user-friendly graphical interface for efficient interaction. This chapter describes the system architecture, design methodology, hardware and software requirements, data flow, implementation process, and user interface design of the proposed system.

3.2 System Architecture

The system follows a simple architecture consisting of input, processing, storage, and output modules.

Architecture Diagram

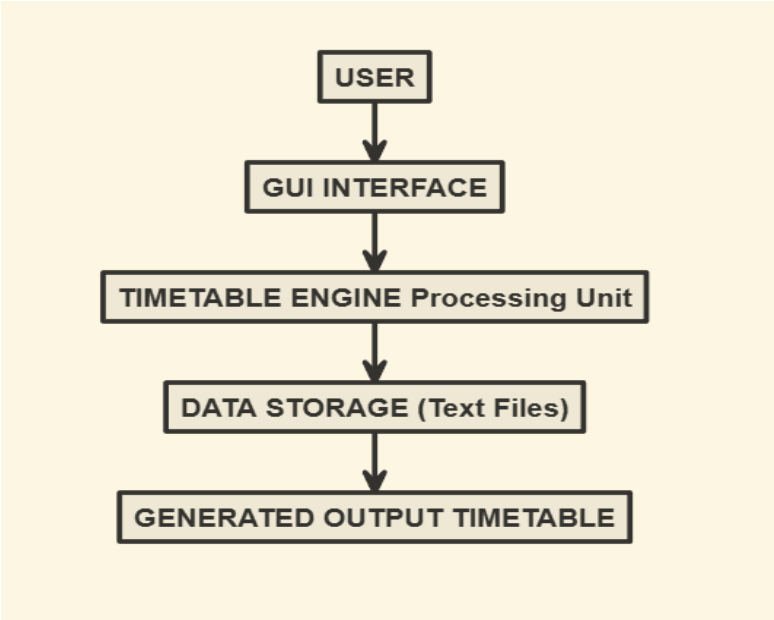


Figure 3.1: Architecture of Proposed System

Description

The user enters subject details through the graphical user interface. The timetable engine processes the entered information and applies scheduling rules to generate a timetable. The generated timetable is displayed to the user and stored in a text file for future reference.

3.3 System Requirements

Hardware Requirements

Table 3.1: Hardware Requirements of the Proposed System

Components	**Specification**
Processor	Intel Core i3 or above
RAM	4 GB Minimum
Storage	500 MB Free Space
Display	1366 × 768 Resolution

Software Requirements

Table 3.2: Software Requirements of the Proposed System

Software	**Version**
Operating system	Windows 10/11
Programming Language	Python
Storage	Text Files
IDE	VS Code

3.4 Methodology

The development of the Automated Timetable Scheduler System follows a structured approach.

Step 1: Data Entry
The user enters subject information through the graphical user interface.

Step 2: Validation
The system validates the entered data and checks for missing information.

Step 3: Processing
The scheduling module processes the subject information and allocates periods according to predefined rules.

Step 4: Timetable Generation
The timetable is generated automatically without scheduling conflicts.

Step 5: Display and Storage
The generated timetable is displayed to the user and stored in a file for future use.

3.5 Flowchart

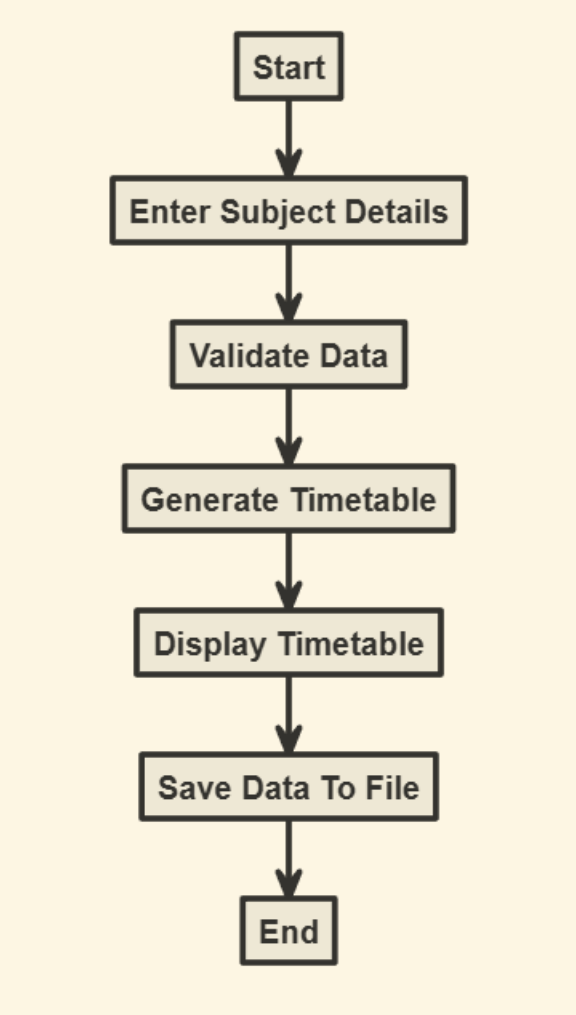


Figure 3.2: System Flowchart

3.6 Data Flow Diagram (DFD)

Level 0 DFD

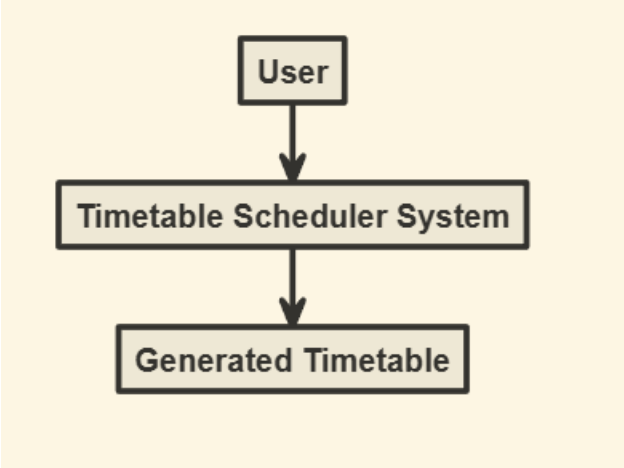


Figure 3.3: Level 0 DFD Flowchart

Level 1 DFD

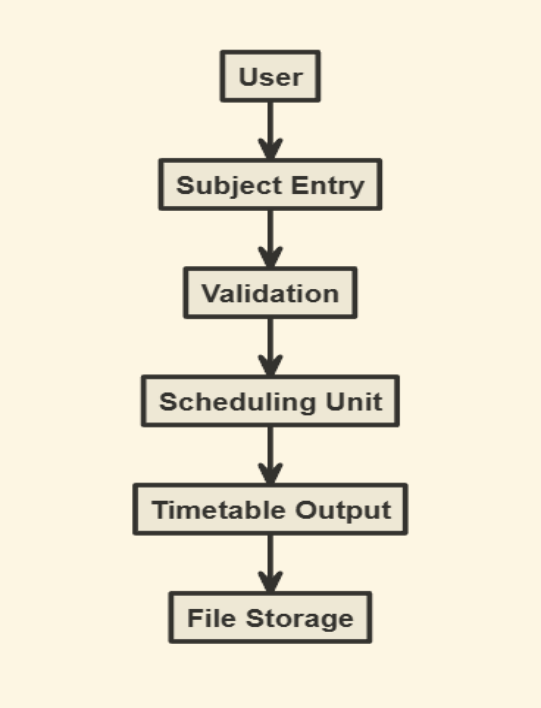


Figure 3.4: Level 1 DFD Flowchart

3.7 Module Description

Subject Management Module

This module allows users to enter and manage subject information. It acts as the primary input source for the timetable generation process.

Timetable Generation Module

This is the core module of the system. It processes subject information and generates a timetable automatically according to predefined scheduling rules.

Display Module

This module displays the generated timetable in a structured format through the graphical interface.

Storage Module

The storage module saves timetable information into text files for future access and modifications.

3.8 User Interface Design

The graphical user interface is designed using the Tkinter library. The interface is simple and user-friendly, enabling users to interact with the system without technical expertise.

Main Features

- Subject Entry Form
- Generate Timetable Button
- Timetable Display Area
- Save Functionality

Screenshot 3.1: Main Window

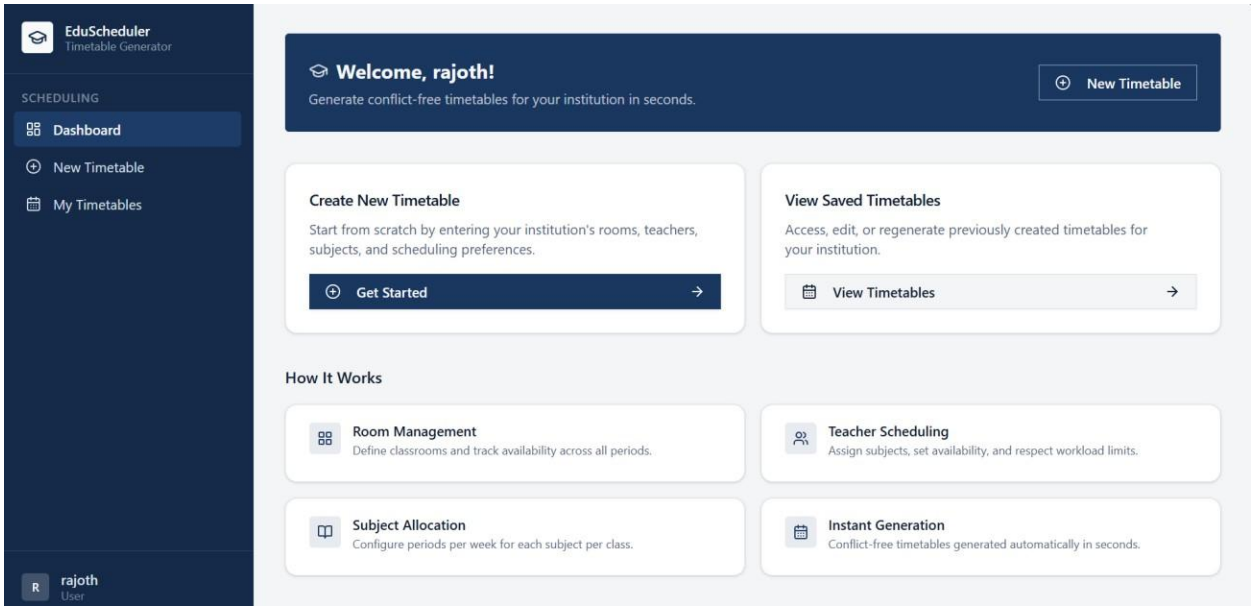


Figure 3.5: Dashboard View

Description:
The main window provides options for entering subject information and generating timetables.

Screenshot 3.3: Class Entry Form

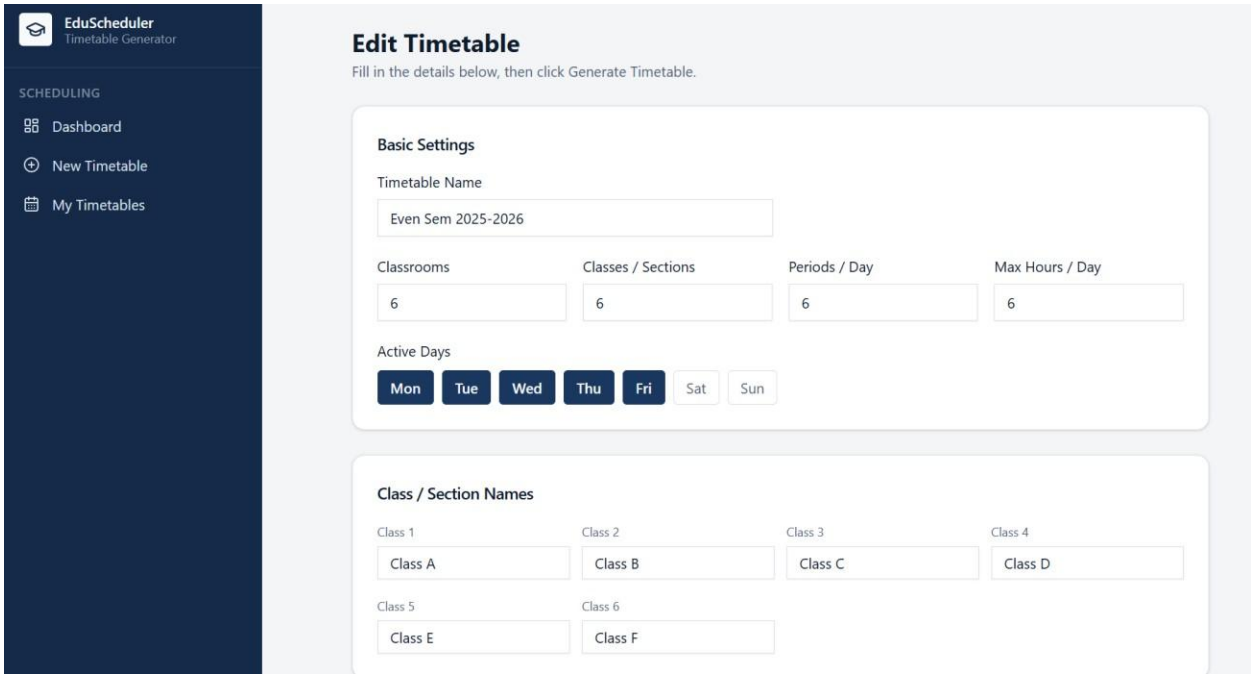


Figure 3.6

Screenshot 3.4: Class Entry Form

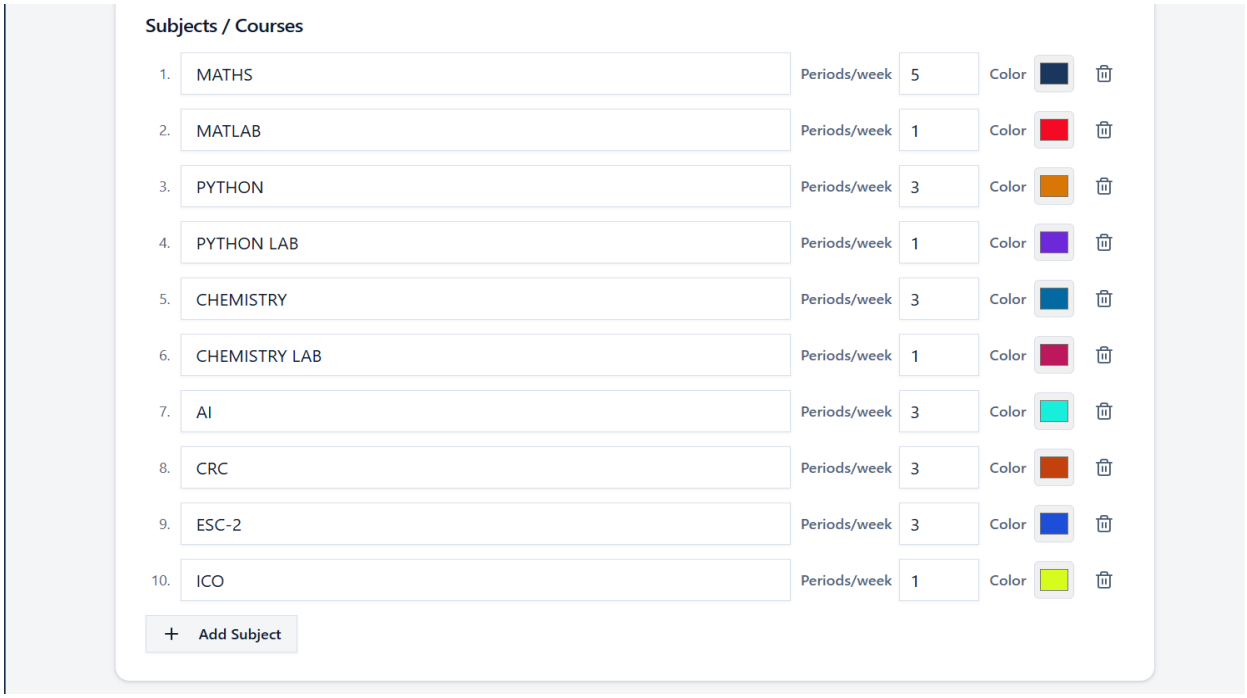


Figure 3.7

Screenshot 3.4: Teacher Entry Form

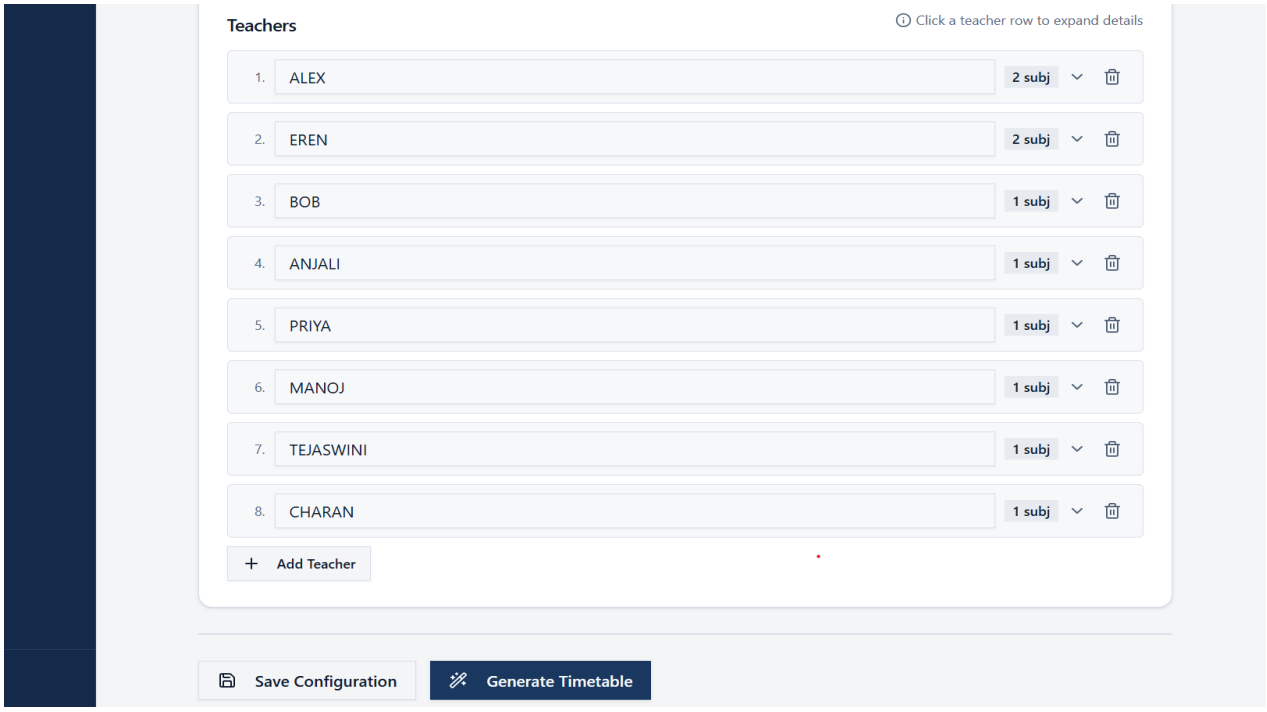


Figure 3.8

Description:
The user enters teacher details through this form.

Screenshot 3.5: Generated Timetable

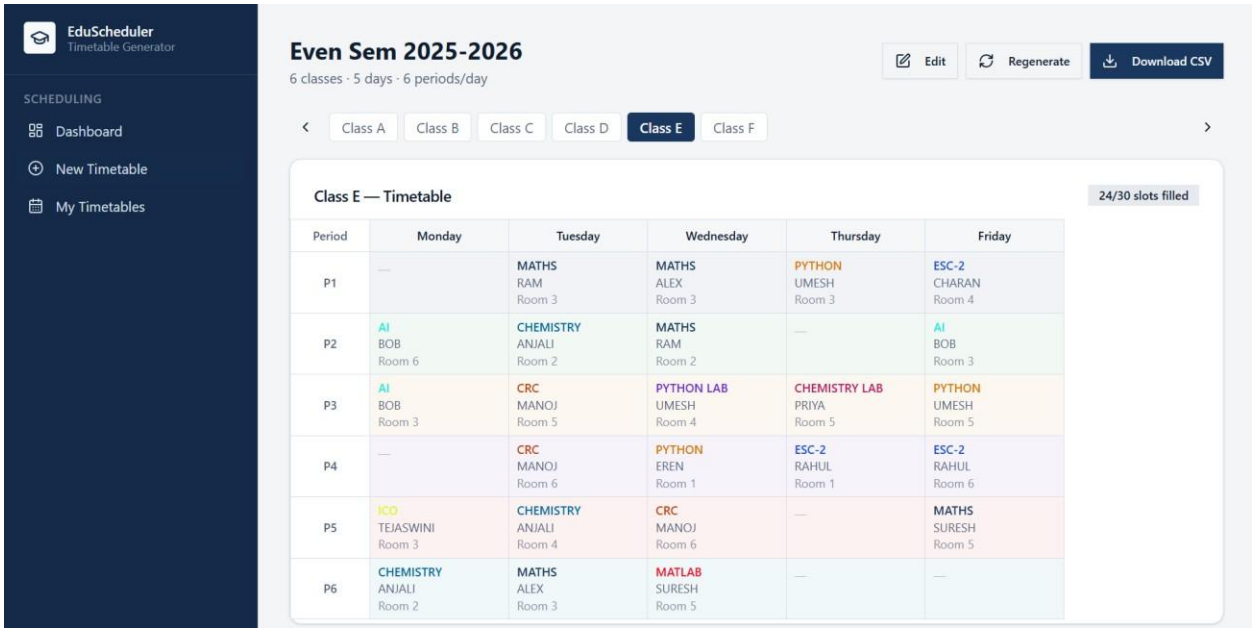


Figure 3.9: Generated Timetable

Description:
This screen displays the automatically generated timetable.

3.9 Codes

```
import tailwindAnimate from 'tailwindcss-animate';
import containerQuery from '@tailwindcss/container-queries';
import intersect from 'tailwindcss-intersect';

export default {
    darkMode: ['class'],
    content: [
        './index.html',
        './pages/**/*.{ts,tsx}',
        './components/**/*.{ts,tsx}',
        './app/**/*.{ts,tsx}',
        './src/**/*.{ts,tsx}',
        './node_modules/streamdown/dist/**/*.js'
    ],
    safelist: ['border', 'border-border'],
    prefix: '',
    theme: {
        container: {
            center: true,
            padding: '2rem',
            screens: {
                '2xl': '1400px'
            }
        },
        extend: {
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                borderColor: {
                    border: 'hsl(var(--border))'
                },
                input: 'hsl(var(--input))',
                ring: 'hsl(var(--ring))',
                background: 'hsl(var(--background))',
```

Figure 3.10

```
                foreground: 'hsl(var(--foreground))',
                primary: {
                    DEFAULT: 'hsl(var(--primary))',
                    foreground: 'hsl(var(--primary-foreground))'
                },
                secondary: {
                    DEFAULT: 'hsl(var(--secondary))',
                    foreground: 'hsl(var(--secondary-foreground))'
                },
                destructive: {
                    DEFAULT: 'hsl(var(--destructive))',
                    foreground: 'hsl(var(--destructive-foreground))'
                },
                muted: {
                    DEFAULT: 'hsl(var(--muted))',
                    foreground: 'hsl(var(--muted-foreground))'
                },
                accent: {
                    DEFAULT: 'hsl(var(--accent))',
                    foreground: 'hsl(var(--accent-foreground))'
                },
                popover: {
                    DEFAULT: 'hsl(var(--popover))',
                    foreground: 'hsl(var(--popover-foreground))'
                },
                card: {
                    DEFAULT: 'hsl(var(--card))',
                    foreground: 'hsl(var(--card-foreground))'
                },
```

Figure 3.11

```
63                education: {
64                    blue: 'hsl(var(--education-blue))',
65                    green: 'hsl(var(--education-green))'
66                },
67                success: 'hsl(var(--success))',
68                warning: 'hsl(var(--warning))',
69                info: 'hsl(var(--info))',
70                sidebar: {
71                    DEFAULT: 'hsl(var(--sidebar-background))',
72                    background: 'hsl(var(--sidebar-background))',
73                    foreground: 'hsl(var(--sidebar-foreground))',
74                    primary: 'hsl(var(--sidebar-primary))',
75                    'primary-foreground': 'hsl(var(--sidebar-primary-foreground))',
76                    accent: 'hsl(var(--sidebar-accent))',
77                    'accent-foreground': 'hsl(var(--sidebar-accent-foreground))',
78                    border: 'hsl(var(--sidebar-border))',
79                    ring: 'hsl(var(--sidebar-ring))'
80                },
81                chart: {
82                    '1': 'hsl(var(--chart-1))',
83                    '2': 'hsl(var(--chart-2))',
84                    '3': 'hsl(var(--chart-3))',
85                    '4': 'hsl(var(--chart-4))',
86                    '5': 'hsl(var(--chart-5))'
87                }
88            },
89            borderRadius: {
90                lg: 'var(--radius)',
91                md: 'calc(var(--radius) - 2px)',
```

Figure 3.12

```
92                sm: 'calc(var(--radius) - 4px)'
93            },
94            backgroundImage: {
95                'gradient-primary': 'var(--gradient-primary)',
96                'gradient-card': 'var(--gradient-card)',
97                'gradient-background': 'var(--gradient-background)'
98            },
99            boxShadow: {
100               card: 'var(--shadow-card)',
101               hover: 'var(--shadow-hover)'
102           },
103           keyframes: {
104               'accordion-down': {
105                   from: {
106                       height: '0'
107                   },
108                   to: {
109                       height: 'var(--radix-accordion-content-height)'
110                   }
111               },
112               'accordion-up': {
113                   from: {
114                       height: 'var(--radix-accordion-content-height)'
115                   },
116                   to: {
117                       height: '0'
118                   }
119               },
120               'fade-in': {
```

Figure 3.11

```
                    from: {
                        opacity: '0',
                        transform: 'translateY(10px)'
                    },
                    to: {
                        opacity: '1',
                        transform: 'translateY(0)'
                    }
                },
                'slide-in': {
                    from: {
                        opacity: '0',
                        transform: 'translateX(-20px)'
                    },
                    to: {
                        opacity: '1',
                        transform: 'translateX(0)'
                    }
                }
            },
            animation: {
                'accordion-down': 'accordion-down 0.2s ease-out',
                'accordion-up': 'accordion-up 0.2s ease-out',
                'fade-in': 'fade-in 0.5s ease-out',
                'slide-in': 'slide-in 0.5s ease-out'
            }
```

Figure 3.14

```
        }
    },
    plugins: [
        tailwindAnimate,
        containerQuery,
        intersect,
        function ({addUtilities}) {
            addUtilities(
                {
                    '.border-t-solid': {'border-top-style': 'solid'},
                    '.border-r-solid': {'border-right-style': 'solid'},
                    '.border-b-solid': {'border-bottom-style': 'solid'},
                    '.border-l-solid': {'border-left-style': 'solid'},
                    '.border-t-dashed': {'border-top-style': 'dashed'},
                    '.border-r-dashed': {'border-right-style': 'dashed'},
                    '.border-b-dashed': {'border-bottom-style': 'dashed'},
                    '.border-l-dashed': {'border-left-style': 'dashed'},
                    '.border-t-dotted': {'border-top-style': 'dotted'},
                    '.border-r-dotted': {'border-right-style': 'dotted'},
                    '.border-b-dotted': {'border-bottom-style': 'dotted'},
                    '.border-l-dotted': {'border-left-style': 'dotted'},
                },
                ['responsive']
            );
        },
    ],
};
```

Figure 3.15

```json
{
  "compilerOptions": {
    "tsBuildInfoFile": "./node_modules/.tmp/tsconfig.app.tsbuildinfo",
    "target": "ES2020",
    "useDefineForClassFields": true,
    "lib": ["ES2020", "DOM", "DOM.Iterable"],
    "module": "ESNext",
    "skipLibCheck": true,
    "resolveJsonModule": true,
    "esModuleInterop": true,
    "moduleResolution": "bundler",
    "allowImportingTsExtensions": true,
    "isolatedModules": true,
    "moduleDetection": "force",
    "noEmit": true,
    "jsx": "react-jsx",
    "strict": true,
    "noUnusedLocals": true,
    "noUnusedParameters": true,
    "noFallthroughCasesInSwitch": true,
    "noUncheckedSideEffectImports": true,
    "baseUrl": ".",
    "paths": {
      "@/*": ["./src/*"]
    },
    "typeRoots": ["./node_modules/**/*"]
  },
  "include": ["src"]
}
```

Figure 3.16

```json
{
  "include": [
    "./src"
  ],
  "exclude": [
    "./src/**/*.test.ts",
    "./src/**/*.spec.ts",
    "./src/components/ui"
  ],
  "compilerOptions": {
    "jsx": "react-jsx",
    "moduleResolution": "bundler",
    "allowImportingTsExtensions": true,
    "target": "ES2020",
    "lib": [
      "ES2020",
      "DOM",
      "DOM.Iterable"
    ],
    "module": "ESNext",
    "skipLibCheck": true,
    "strictNullChecks": true,
    "noEmit": true,
    "paths": {
      "@/*": [
        "./src/*"
      ]
    }
  }
}
```

Figure 3.17



```
{
  "compilerOptions": {
    "target": "ES2020",
    "useDefineForClassFields": true,
    "lib": [
      "ES2020",
      "DOM",
      "DOM.Iterable"
    ],
    "module": "ESNext",
    "skipLibCheck": true,
    "baseUrl": ".",
    "paths": {
      "@/*": [
        "./src/*"
      ]
    }
  },
  "files": [],
  "references": [
    {
      "path": "./tsconfig.app.json"
    },
    {
      "path": "./tsconfig.node.json"
    }
  ]
}
```

Figure 3.18



```
{
  "compilerOptions": {
    "tsBuildInfoFile": "./node_modules/.tmp/tsconfig.node.tsbuildinfo",
    "target": "ES2022",
    "lib": ["ES2023"],
    "module": "ESNext",
    "skipLibCheck": true,
    "moduleResolution": "bundler",
    "allowImportingTsExtensions": true,
    "isolatedModules": true,
    "moduleDetection": "force",
    "noEmit": true,
    "strict": true,
    "noUnusedLocals": true,
    "noUnusedParameters": true,
    "noFallthroughCasesInSwitch": true,
    "noUncheckedSideEffectImports": true
  },
  "include": ["vite.config.ts"]
}
```

Figure 3.19

```
7   export async function getProfile(userId: string): Promise<Profile | null> {
8     const { data, error } = await supabase
9       .from('profiles')
10      .select('*')
11      .eq('id', userId)
12      .maybeSingle();
13
14    if (error) {
15      console.error('Failed to load profile:', error);
16      return null;
17    }
18    return data;
19  }
20  interface AuthContextType {
21    user: User | null;
22    profile: Profile | null;
23    loading: boolean;
24    signInWithUsername: (username: string, password: string) => Promise<{ error: Error | null }>;
25    signUpWithUsername: (username: string, password: string) => Promise<{ error: Error | null }>;
26    signOut: () => Promise<void>;
27    refreshProfile: () => Promise<void>;
28  }
29
30  const AuthContext = createContext<AuthContextType | undefined>(undefined);
31
32  export function AuthProvider({ children }: { children: ReactNode }) {
33    const [user, setUser] = useState<User | null>(null);
34    const [profile, setProfile] = useState<Profile | null>(null);
35    const [loading, setLoading] = useState(true);
36
37    const refreshProfile = async () => {
38      if (!user) {
```

Figure 3.20

```
40        return;
41      }
42
43      const profileData = await getProfile(user.id);
44      setProfile(profileData);
45    };
46
47    useEffect(() => {
48      supabase
49        .auth
50        .getSession()
51        .then(({ data: { session } }) => {
52          setUser(session?.user ?? null);
53          if (session?.user) {
54            getProfile(session.user.id).then(setProfile);
55          }
56        })
57        // @ts-ignore
58        .catch(error => {
59          toast.error(`Failed to load session: ${error.message}`);
60        })
61        .finally(() => {
62          setLoading(false);
63        });
64
65      // In this function, do NOT use any await calls. Use `.then()` instead to avoid deadlocks.
66      const { data: { subscription } } = supabase.auth.onAuthStateChange((_event, session) => {
67        setUser(session?.user ?? null);
68        if (session?.user) {
69          getProfile(session.user.id).then(setProfile);
70        } else {
```

Figure 3.21

```
    }
  });

  return () => subscription.unsubscribe();
}, []);

const signInWithUsername = async (username: string, password: string) => {
  try {
    const email = `${username}@miaoda.com`;
    const { error } = await supabase.auth.signInWithPassword({ email, password });
    if (error) throw error;
    return { error: null };
  } catch (error) {
    return { error: error as Error };
  }
};

const signUpWithUsername = async (username: string, password: string) => {
  try {
    const email = `${username}@miaoda.com`;
    const { data, error } = await supabase.auth.signUp({ email, password });
    if (error) throw error;
    // Sync username into profiles
    if (data.user) {
      await supabase
        .from('profiles')
        .update({ username })
        .eq('id', data.user.id);
    }
    return { error: null };
  } catch (error) {
```

Figure 3.22

```
    return { error: null };
  } catch (error) {
    return { error: error as Error };
  }
};

const signOut = async () => {
  await supabase.auth.signOut();
  setUser(null);
  setProfile(null);
};

return (
  |
}

export function useAuth() {
  const context = useContext(AuthContext);
  if (context === undefined) {
    throw new Error('useAuth must be used within an AuthProvider');
  }
  return context;
}
```

Figure 3.23

3.10 Implementation Details

The project is implemented using Python programming language due to its simplicity and extensive library support. Tkinter is used to create the graphical user interface. The application consists of multiple functions responsible for data input, validation, timetable generation, display, and storage.

The timetable generation algorithm processes subject information and assigns periods systematically to avoid scheduling conflicts. Data is stored using text file handling techniques, allowing the generated timetable to be retrieved when needed.

The modular structure of the program improves maintainability and simplifies future enhancements.

3.11 Summary

This chapter presented the design and implementation details of the Automated Timetable Scheduler System. The system architecture, methodology, flowcharts, data flow diagrams, module descriptions, and implementation details were discussed. The developed system provides an efficient and user-friendly solution for automated timetable generation.

CHAPTER-4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from the implementation and testing of the Automated Timetable Scheduler System. The primary objective of the system is to automate the timetable generation process and reduce the difficulties associated with manual scheduling. Various test cases were conducted to evaluate the performance, accuracy, and reliability of the developed system. The results obtained from these tests are analyzed and discussed in detail to determine the effectiveness of the proposed solution.

4.2 Results Analysis

The Automated Timetable Scheduler System was tested using different sets of subject information to verify its functionality and performance. The system successfully accepted user inputs, processed subject details, and generated timetables without scheduling conflicts. The generated timetables were displayed correctly through the graphical user interface and stored successfully in text files.

During testing, the application demonstrated consistent performance in generating timetables within a short period of time. Compared to manual timetable preparation, the automated system significantly reduced the effort required to create schedules. The timetable generation process was completed instantly after entering the required subject information, thereby improving overall efficiency.

The system was also tested with different numbers of subjects to evaluate its capability to handle varying input sizes. The results indicated that the application maintained its performance and generated valid schedules regardless of the number of subjects entered. No major errors or system failures were observed during testing.

Test Case Analysis

Table:4.1: Test Case

Test Case	Input Condition	Expected Result	Obtained Result	Status
TC-01	Valid subject details	Subject data accepted	Subject data accepted successfully	Pass
TC-02	Generate timetable	Timetable generated	Timetable generated successfully	Pass
TC-03	Empty input field	Error message displayed	Error message displayed correctly	Pass
TC-04	Multiple subjects entered	Conflict-free timetable	Conflict-free timetable generated	Pass
TC-05	Save timetable	Timetable stored in file	Timetable stored successfully	Pass

The test results confirm that all major functionalities of the system operate according to the specified requirements.

4.3 Discussion

The implementation of the Automated Timetable Scheduler System demonstrates the advantages of automating academic scheduling activities. Traditional timetable preparation methods require substantial manual effort and are often prone to errors. The developed system eliminates many of these challenges by automating the scheduling process and ensuring accurate timetable generation.

One of the significant observations during testing was the reduction in timetable preparation time. Manual timetable generation may require several hours depending on the number of subjects and scheduling constraints. In contrast, the developed system generates timetables within seconds after receiving the required input data.

The graphical user interface contributed significantly to the usability of the application. Users were able to enter subject information easily and generate schedules without requiring specialized technical knowledge. The interface design improved accessibility and enhanced user experience.

The storage functionality also proved beneficial as it allowed generated timetables to be saved and retrieved whenever necessary. This feature improves convenience and reduces the need for repeated timetable generation.

Although the system performed effectively, certain limitations were identified. The current version focuses only on subject scheduling and does not consider classroom allocation, faculty scheduling, or examination timetable generation. These limitations provide opportunities for future enhancements.

4.4 Validation

Validation was performed to verify that the developed system satisfies the objectives defined at the beginning of the project. The generated timetables were compared with manually prepared schedules to evaluate accuracy and efficiency.

The comparison showed that the automated system produced valid and conflict-free timetables while requiring significantly less time and effort. Furthermore, the system maintained consistency in timetable generation, reducing the possibility of human errors that commonly occur in manual scheduling processes.

Comparison Between Manual and Automated Timetable Generation

Table:4.2: Comparison Between Manual Timetable Preparation and Automated Timetable Scheduler System

Parameter	Manual Method	Automated System
Time Required	High	Very Low
Human Effort	High	Low
Accuracy	Moderate	High
Scheduling Conflicts	Possible	Minimized
Ease of Modification	Moderate	Easy
Efficiency	Moderate	High

The validation results indicate that the proposed system successfully achieves its intended objectives. The generated timetables are accurate, reliable, and suitable for practical use in educational institutions.

4.5 Performance Evaluation

The performance of the Automated Timetable Scheduler System was evaluated based on several criteria, including accuracy, speed, usability, and reliability.

The system achieved high accuracy in timetable generation and consistently produced schedules without conflicts. The processing time was minimal, enabling rapid timetable creation even when

multiple subjects were entered. The graphical user interface simplified user interaction and improved overall usability. Additionally, the system demonstrated reliable operation throughout the testing process without significant failures or unexpected behavior.

These performance characteristics confirm that the system is capable of providing an effective solution for academic timetable generation.

4.6 Summary

This chapter presented the results obtained from testing and evaluating the Automated Timetable Scheduler System. The analysis demonstrated that the system successfully generates conflict-free timetables while reducing the time and effort associated with manual scheduling. The discussion highlighted the benefits of automation, improved efficiency, and enhanced usability. Validation results confirmed that the developed system meets the project objectives and provides a practical solution for timetable generation in educational institutions.

CHAPTER-5

CONCLUSION

5.1 Introduction

The Automated Timetable Scheduler System was developed with the objective of simplifying and automating the timetable preparation process in educational institutions. Timetable scheduling is a critical administrative task that requires careful allocation of subjects across available time slots. Traditional timetable preparation methods often involve significant manual effort and are susceptible to scheduling conflicts and human errors.

The proposed system provides an automated solution that enables educational institutions to generate timetables quickly and accurately. Through the use of Python programming language and the Tkinter graphical user interface library, a user-friendly application was developed to automate scheduling activities. This chapter summarizes the outcomes of the project, discusses its achievements and limitations, and presents recommendations and future enhancement opportunities.

5.2 Summary of Findings

The study conducted during the development of the Automated Timetable Scheduler System revealed several important observations regarding timetable preparation and scheduling processes.

The analysis of existing scheduling methods indicated that manual timetable preparation is a time-consuming and repetitive task. Educational institutions often spend considerable time arranging subjects and ensuring that schedules are free from conflicts. As the number of subjects increases, managing timetables becomes increasingly complex and difficult.

The developed system successfully addressed these challenges by automating timetable generation. During testing, the application demonstrated the ability to generate accurate schedules based on the subject information entered by users. The generated timetables were organized systematically and displayed in a format that was easy to understand and use.

The findings also revealed that automation significantly reduces administrative workload. The timetable generation process that traditionally requires hours of manual effort can be completed within seconds using the developed system. Furthermore, the possibility of scheduling errors is greatly minimized because the system follows predefined scheduling rules consistently.

The implementation of a graphical user interface improved user interaction and made the application accessible even to users with limited technical knowledge. Overall, the project successfully demonstrated the practical benefits of software-based timetable automation.

5.3 Achievements of the Project

The development and implementation of the Automated Timetable Scheduler System resulted in several significant achievements.

One of the major achievements was the successful automation of timetable generation. The system can generate schedules automatically without requiring manual arrangement of subjects. This achievement directly contributes to reducing administrative workload and improving operational efficiency.

Another important achievement was the development of a user-friendly graphical interface. The interface allows users to enter subject details, generate timetables, and view results easily. The simplicity of the interface ensures that users can operate the system without extensive training.

The project also achieved successful integration of data validation and file storage functionalities. Input validation ensures that only valid data is processed, while file handling techniques allow generated timetables to be saved for future reference.

The testing phase confirmed that the system performs reliably under different conditions. Various subject combinations were tested, and the system consistently generated valid timetables without major errors or failures.

In addition, the project provided valuable experience in software design, programming, testing, and documentation. The knowledge gained during development contributes to a deeper understanding of automation technologies and their applications in educational administration.

5.4 Benefits of the Proposed System

The Automated Timetable Scheduler System offers numerous benefits to educational institutions.

The system significantly reduces the time required for timetable preparation. Instead of manually arranging schedules, administrators can generate timetables automatically within a few seconds. This saves valuable time and allows staff members to focus on other important academic responsibilities.

The application also improves accuracy by minimizing human errors. Since the timetable is generated according to predefined scheduling rules, the chances of conflicts and incorrect allocations are substantially reduced.

Another benefit is improved consistency. Manual scheduling may produce different results depending on the individual preparing the timetable. In contrast, the automated system follows a consistent procedure every time it generates a schedule.

The system is cost-effective because it is developed using open-source technologies such as Python and Tkinter. Educational institutions can implement the solution without significant financial investment.

Furthermore, the user-friendly design improves accessibility and ensures that users can operate the system efficiently regardless of their technical expertise.

5.5 Limitations of the System

Although the Automated Timetable Scheduler System successfully fulfills its intended objectives, certain limitations exist in the current version.

The system focuses exclusively on subject scheduling and does not include faculty allocation. In real-world educational environments, faculty availability often plays an important role in timetable preparation. The current system does not consider such constraints.

Similarly, classroom allocation is not included in the scheduling process. Educational institutions typically need to assign classrooms based on subject requirements and room availability. This functionality remains outside the scope of the current implementation.

The application uses text files for data storage rather than a database management system. While this approach is sufficient for small-scale implementations, larger institutions may require more advanced storage solutions to manage extensive scheduling data.

The system is designed as a standalone desktop application and does not support online access. Users must operate the software on a local computer, limiting accessibility and collaboration.

Finally, the scheduling logic is based on predefined rules and does not incorporate advanced optimization techniques such as Artificial Intelligence or Genetic Algorithms. Consequently, there may be opportunities for further optimization in complex scheduling scenarios.

5.6 Recommendations

Several recommendations can be made to enhance the effectiveness and functionality of the system.

A database management system such as MySQL or SQLite can be integrated to improve data storage and retrieval capabilities. Database integration would provide better scalability and enable efficient management of large datasets.

Additional scheduling constraints can also be incorporated to support faculty allocation, classroom assignment, and workload balancing. These enhancements would make the system more suitable for real-world institutional environments.

The graphical user interface can be further improved by adding advanced features such as timetable editing, searching, filtering, and printing. Such functionalities would increase convenience and flexibility for users.

Security features such as user authentication and role-based access control can also be implemented to protect timetable information from unauthorized access.

Regular maintenance and updates should be performed to ensure that the system remains reliable and compatible with future technological developments.

5.7 Future Scope

The future scope of the Automated Timetable Scheduler System is extensive. Several advanced technologies and features can be incorporated to transform the application into a comprehensive academic management solution.

One potential enhancement is the development of a web-based version of the system. A web application would allow users to access timetable information from any location through an internet connection. This would improve accessibility and facilitate collaborative scheduling.

Another important enhancement is the integration of mobile applications. Mobile support would enable students and faculty members to view timetables directly from their smartphones.

Artificial Intelligence and Machine Learning techniques can also be incorporated to optimize timetable generation. Intelligent algorithms can analyze scheduling patterns and automatically generate highly efficient schedules while satisfying multiple constraints.

Future versions of the system may include faculty scheduling, classroom management, examination timetable generation, attendance integration, and notification systems. These features would extend the functionality of the application and provide a more comprehensive educational management platform.

Cloud-based storage and synchronization can further improve accessibility and data security. Institutions would be able to store timetable information securely while providing real-time updates to users.

The implementation of these enhancements would significantly increase the capabilities and usefulness of the Automated Timetable Scheduler System.

5.8 Conclusion

The Automated Timetable Scheduler System successfully demonstrates the application of software technology in solving academic scheduling problems. The project achieved its primary objective of automating timetable generation and reducing the difficulties associated with manual scheduling.

Through proper planning, design, implementation, and testing, the system was able to generate conflict-free timetables efficiently and accurately. The user-friendly graphical interface, reliable performance, and effective scheduling functionality contribute to the overall success of the project.

The findings of this study indicate that automated scheduling systems can significantly improve administrative efficiency, reduce workload, and minimize human errors. Although certain limitations exist, the proposed system provides a strong foundation for future enhancements and advanced scheduling solutions.

In conclusion, the Automated Timetable Scheduler System represents a practical, efficient, and cost-effective approach to academic timetable generation and has the potential to evolve into a comprehensive educational management system through future developments.

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